

Study on the Effect of Different Occlusal Schemes on Temporomandibular Joint Health in Complete Denture Wearers

Ravi Ranjan Sinha¹, Poda Alekhya¹, Srinu G¹, Harprit Kaur¹, Pattar Neelavathi¹, Carrolene Langpoklakpam²

¹Department of Prosthodontics and Crown & Bridge, The Oxford Dental College, Bengaluru, Karnataka, India,
²Department of Pediatric and Preventive Dentistry, Manav Rachna Dental College, SDS, MRIIRS, Faridabad, Haryana, India

Submitted: 14-Jul-2025
 Revised: 20-Jul-2025
 Accepted: 10-Aug-2025
 Published: 17-Dec-2025

INTRODUCTION

The temporomandibular joint (TMJ) plays a vital role in mastication, speech, and overall oral function. In edentulous patients, the loss of natural teeth leads to altered mandibular movements and changed biomechanics, potentially predisposing them to temporomandibular disorders (TMDs).^[1] Complete denture prostheses aim to restore these functions, and the occlusal scheme adopted during fabrication can significantly influence the health of the TMJ.^[2]

Different occlusal schemes have been developed to improve denture stability, masticatory efficiency, and patient comfort. Among them, balanced occlusion

ABSTRACT **Background:** Temporomandibular joint (TMJ) disorders are commonly observed in edentulous patients, potentially exacerbated by improper occlusal schemes in complete dentures. Various occlusal concepts, including balanced occlusion, lingualized occlusion, and monoplane occlusion, are used in prosthodontics, each claiming advantages in preserving TMJ health. **Materials and Methods:** A total of 60 completely edentulous patients aged between 50 and 75 years were selected and randomly divided into three groups ($n = 20$) based on the occlusal scheme: Group A – Balanced Occlusion, Group B – Lingualized Occlusion, and Group C – Monoplane Occlusion. All patients received new maxillary and mandibular complete dentures fabricated using conventional techniques. TMJ health was assessed at baseline, 3 months, and 6 months post-insertion using Helkimo's Clinical Dysfunction Index and patient-reported outcomes on pain and joint sounds. **Results:** At the 6-month follow-up, Group A showed minimal TMJ symptoms with a mean dysfunction index score of 0.8 ± 0.5 , compared to 1.2 ± 0.6 in Group B and 2.1 ± 0.7 in Group C. Joint sounds and pain were reported in 10% of patients in Group A, 15% in Group B, and 35% in Group C. Statistical analysis revealed significant differences between the groups ($P < 0.05$), favoring balanced occlusion for TMJ stability. **Conclusion:** Balanced occlusion in complete dentures appears to better maintain TMJ health compared to lingualized and monoplane schemes. It minimizes joint dysfunction symptoms and improves patient comfort over time.

KEYWORDS: *Balanced occlusion, complete dentures, lingualized occlusion, monoplane occlusion, occlusal schemes, temporomandibular joint, TMJ dysfunction*

is traditionally advocated to provide simultaneous bilateral contact during eccentric movements, potentially distributing occlusal forces evenly and minimizing strain on the TMJ.^[3] Lingualized occlusion, characterized by dominant maxillary lingual cusps occluding in the central fossae of mandibular teeth, aims to combine esthetics with functional balance.^[4] Monoplane occlusion, which uses flat teeth and minimal vertical overlap, is often

Address for correspondence: Dr. Ravi Ranjan Sinha, Department of Prosthodontics and Crown & Bridge, The Oxford Dental College, Bengaluru - 560 068, Karnataka, India. E-mail: r.ravi.sinha@gmail.com

This is an open access article distributed under the terms of the Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License (CC BY-NC-ND), where it is permissible to download and share the work provided it is properly cited. The work cannot be changed in any way or used commercially without permission from the journal.

For reprints contact: WKHLRPMedknow_reprints@wolterskluwer.com

How to cite this article: Sinha RR, Alekhya P, Srinu G, Kaur H, Neelavathi P, Langpoklakpam C. Study on the effect of different occlusal schemes on temporomandibular joint health in complete denture wearers. J Pharm Bioall Sci 2025;17:S3162-4.

Access this article online	
Quick Response Code: 	Website: https://journals.lww.com/jpbs DOI: 10.4103/jpbs.jpbs_1236_25

recommended in patients with resorbed ridges but may compromise TMJ loading due to reduced proprioception and occlusal guidance.^[5]

Although each occlusal scheme has its proponents, the relationship between these schemes and TMJ health in complete denture wearers remains inadequately explored. Previous studies have focused primarily on retention, stability, and patient satisfaction, with limited data on TMJ-related outcomes.^[6,7]

MATERIALS AND METHODS

A total of 60 completely edentulous patients aged between 50 and 75 years were selected based on the following criteria:

- Inclusion: Patients with well-formed, non-flabby ridges and no history of TMJ disorders or neuromuscular diseases.
- Exclusion: Individuals with parafunctional habits (e.g., bruxism), systemic conditions affecting joint function, or prior denture experience within the last five years.

Study groups

Patients were randomly assigned to three groups ($n = 20$ each) based on the occlusal scheme used in their dentures:

- **Group A:** Balanced occlusion
- **Group B:** Lingualized occlusion
- **Group C:** Monoplane occlusion.

Denture fabrication

Complete dentures for all participants were fabricated following standard clinical procedures. Maxillary and mandibular dentures were constructed using heat-cured acrylic resin. Jaw relation records were obtained using wax rims, and teeth were arranged as per the assigned occlusal scheme:

- Balanced occlusion used anatomic teeth with bilateral simultaneous contact.
- Lingualized occlusion incorporated maxillary lingual cusp contacts on mandibular central fossae.
- Monoplane occlusion utilized flat occlusal surfaces without cusp interdigitation.

Denture insertion was followed by post-insertion adjustments over the first week.

TMJ assessment

Temporomandibular joint status was evaluated using:

1. **Helkimo's Clinical Dysfunction Index** – assessing joint sounds, tenderness, limitation of movement, and pain on function.
2. **Patient-reported outcomes** – subjective feedback on joint discomfort and functional issues, collected via a standardized questionnaire.

Evaluations were performed at three time points: baseline (prior to denture insertion), 3 months, and 6 months post-insertion.

Statistical analysis

Data were entered into Microsoft Excel and analyzed using SPSS version 25.0.

RESULTS

A total of 60 participants completed the study, with an equal distribution across the three occlusal scheme groups. All patients were followed for a period of six months post denture insertion. The results are presented in terms of Helkimo's Clinical Dysfunction Index scores and patient-reported symptoms.

Helkimo's clinical dysfunction index scores

At baseline, all three groups showed minimal TMJ dysfunction. Over time, differences emerged in the severity of TMJ-related symptoms. Group A (Balanced occlusion) consistently showed the lowest dysfunction scores, followed by Group B (Lingualized occlusion), while Group C (Monoplane occlusion) recorded the highest scores at all follow-up intervals [Table 1].

Patient-Reported TMJ symptoms

Subjective complaints of joint pain, clicking sounds, and restricted jaw movement were recorded at 6 months. Group A had the fewest complaints, while Group C exhibited a higher frequency of symptoms [Table 2].

Overall, balanced occlusion showed superior results in maintaining TMJ health over 6 months, both objectively [Table 1] and subjectively [Table 2].

DISCUSSION

Balanced occlusion aims to provide bilateral, simultaneous contact during eccentric movements, thereby ensuring even distribution of occlusal forces and

Table 1: Mean Helkimo's Dysfunction Index Scores at Different Time Intervals

Time point	Group A (balanced)	Group B (lingualized)	Group C (monoplane)
Baseline	0.6±0.4	0.7±0.3	0.6±0.5
3 months	0.9±0.5	1.3±0.6	1.9±0.7
6 months	1.1±0.6	1.5±0.7	2.4±0.8

Table 2: Frequency of patient-reported TMJ symptoms at 6 months

Symptom	Group A (%)	Group B (%)	Group C (%)
Joint pain	2 (10%)	3 (15%)	7 (35%)
Clicking/joint sounds	1 (5%)	3 (15%)	6 (30%)
Limited mouth opening	1 (5%)	2 (10%)	5 (25%)

minimizing displacing forces on the denture base.^[1] This biomechanical stability is believed to reduce strain on the TMJ, which aligns with our findings of lower Helkimo index scores and fewer symptoms in Group A. These results are consistent with previous reports suggesting that balanced articulation helps maintain neuromuscular harmony in edentulous patients.^[2,3]

Lingualized occlusion, which seeks to balance esthetics and function, showed intermediate outcomes. While it was slightly less favorable than balanced occlusion in our study, its design—which centralizes occlusal forces on the mandibular ridge—may still offer benefits in cases of moderate ridge resorption.^[4] Some studies support its use for improving esthetics and ease of fabrication, though its long-term effects on TMJ health remain less definitive.^[5]

In contrast, monoplane occlusion, despite being easier to establish and advantageous in cases with severely resorbed ridges or poor neuromuscular control, showed the highest levels of TMJ dysfunction in this study. The flat occlusal surfaces and absence of cusp-guided movements may reduce proprioceptive feedback and occlusal stability, potentially contributing to muscle incoordination and joint strain.^[6] Prior literature has indicated that such schemes may compromise masticatory efficiency and functional adaptation, particularly in patients with high functional demands.^[7]

It is also important to recognize the role of individual adaptability and neuromuscular coordination in how patients respond to different occlusal designs. While occlusal scheme selection is important, patient education, regular follow-up, and timely denture adjustments also play a crucial role in maintaining TMJ health.^[8]

CONCLUSION

Balanced occlusion in complete dentures appears to better maintain TMJ health than lingualized and monoplane schemes. It minimizes joint dysfunction symptoms and improves patient comfort over time.

Financial support and sponsorship

Nil.

Conflicts of interest

There are no conflicts of interest.

REFERENCES

1. Dervis E. Changes in temporomandibular disorders after treatment with new complete dentures. *J Oral Rehabil* 2004;31:320–6.
2. Yannikakis S, Zissis A, Harrison A. The prevalence of temporomandibular disorders among two different denture-wearing populations. *Eur J Prosthodont Restor Dent* 2009;17:35–40.
3. Zuccolotto MC, Vitti M, Nóbilo KA, Regalo SC, Siéssere S, Bataglioni C. Electromyographic evaluation of masseter and anterior temporalis muscles in rest position of edentulous patients with temporomandibular disorders, before and after using complete dentures with sliding plates. *Gerodontology* 2007;24:105–10.
4. Gray RJ, McCord JF, Murtaza G, Siddique M. The incidence of temporomandibular disorder signs in patients wearing complete dentures compared to patients with a natural dentition. *Eur J Prosthodont Restor Dent* 1997;5:99–103.
5. Bordin TB, Conci RA, Pezzini MM, Pezzini RP, Mendonça MJ. Prevalence of signs and symptoms of temporomandibular disorders (TMD) in patients wearing bimaxillary complete dentures, removable partial dentures and in students with natural dentition. *Acta Odontol Latinoam* 2013;26:173–80.
6. Köhler AA, Hugoson A, Magnusson T. Clinical signs indicative of temporomandibular disorders in adults: Time trends and associated factors. *Swed Dent J* 2013;37:1–11.
7. Dabadie M. [Joint and muscular pathology before, during and after treatment of total edentulousness]. *Chir Dent Fr* 1989;59:45–58.
8. Lundeen TF, Scruggs RR, McKinney MW, Daniel SJ, Levitt SR. TMD symptomatology among denture patients. *J Craniomandib Disord* 1990;4:40–6.